AI-generated · Unofficial practice

Biology Mock

Paper 1 (Section A Multiple-choice · Section B Structured)

150 minutes（2½ hours）

FOR REVISION

For revision and self-assessment only. Not an official examination paper.

Difficulty rating：**4.1** / 5.0

INSTRUCTIONS

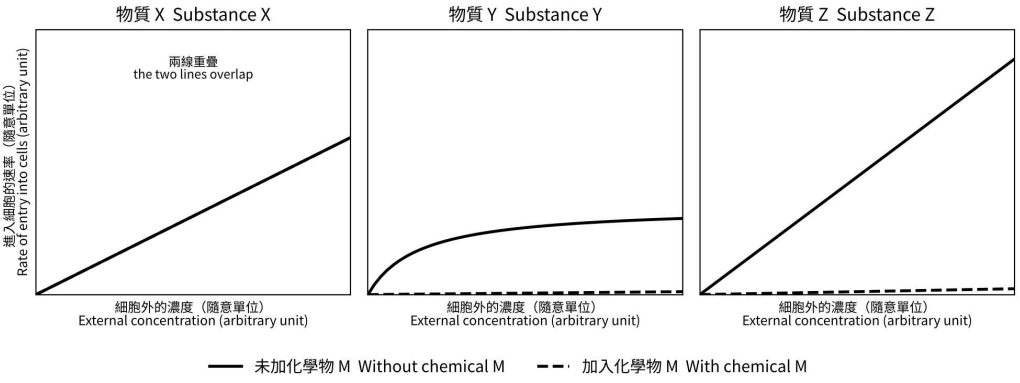
1. This paper consists of Section A (36 multiple-choice questions) and Section B (structured questions and an essay question).
2. Answer ALL questions.
3. Each question in Section A has only ONE correct answer and carries 1 mark.
4. In Section B, write your answers in the spaces provided. Answers should be written in paragraph form where appropriate.
5. The last question in Section B is an essay. Criteria for marking include relevant content, logical presentation and clarity of expression (3 marks for effective communication).
6. The diagrams in this paper are not necessarily drawn to scale.

Independent, unofficial mock paper. Not affiliated with any official examination authority.

Section A Multiple-choice Questions

1. The rates of entry of three substances X, Y and Z into cultured human epithelial cells were investigated. Chemical M binds to all proteins in the cell membrane and makes them non-functional. The results are shown below.

Which of the following combinations correctly identifies substances X, Y and Z?



A. X: chloride ions; Y: glucose; Z: carbon dioxide

B. X: carbon dioxide; Y: chloride ions; Z: glucose

C. X: carbon dioxide; Y: glucose; Z: chloride ions

D. X: glucose; Y: carbon dioxide; Z: chloride ions

2. The table below shows the base composition of one strand (strand 1) of a double-stranded DNA molecule.

Which of the following deductions are correct?

(1) In the strand complementary to strand 1, C makes up 30%.

(2) In the whole DNA molecule, A makes up 26%.

(3) In the whole DNA molecule, G and C together make up 50%.

某 DNA 分子其中一條鏈（鏈 1）的鹼基組成
Base composition of one strand (strand 1) of a DNA molecule

	A (%)	G (%)	C (%)	T (%)
鏈 1 Strand 1	22	30	18	30

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

3. An animal has the genotype AaBb. The two pairs of genes are on different pairs of homologous chromosomes. Assuming no crossing over, the animal produces four types of gametes, AB, Ab, aB and ab, in equal proportions.

Which of the following combinations correctly shows the process that gives the four types of gametes in equal proportions (X), and the stage at which alleles A and a separate (Y)?

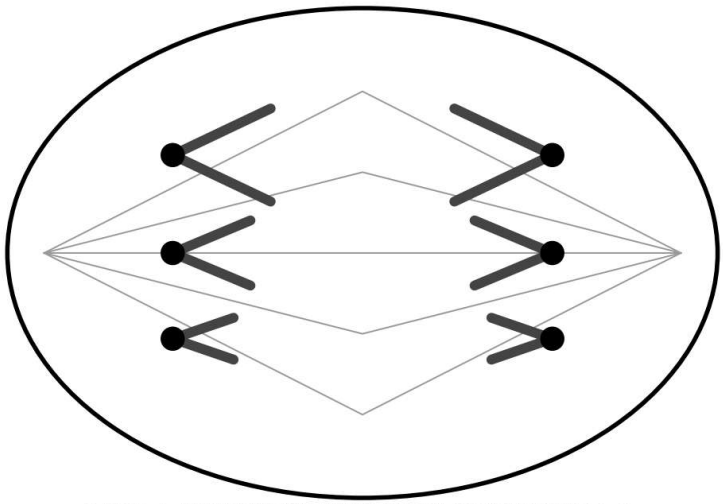
A. X: separation of chromatids in the second meiotic cell division; Y: anaphase of the second meiotic cell division

B. X: random arrangement of each pair of homologous chromosomes at the equator at metaphase of the first meiotic cell division; Y: anaphase of the first meiotic cell division

C. X: random fusion of gametes at fertilisation; Y: anaphase of the first meiotic cell division

D. X: random arrangement of each pair of homologous chromosomes at the equator at metaphase of the first meiotic cell division; Y: anaphase of the second meiotic cell division

Directions: Questions 4 and 5 refer to the diagram below, which shows a cell of an animal undergoing meiotic cell division.



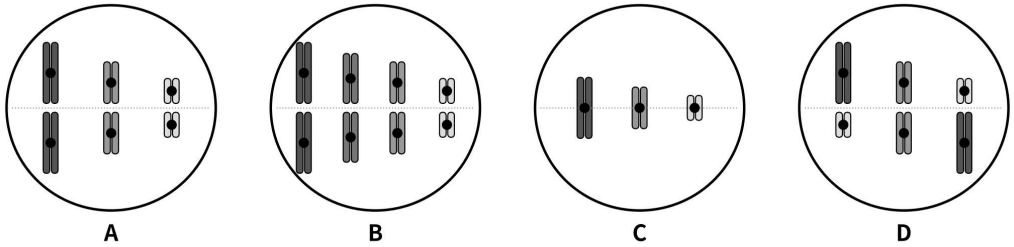
下圖顯示一動物細胞正進行減數細胞分裂（紡錘絲以灰線表示）
Animal cell undergoing meiotic cell division (grey lines: spindle fibres)

4. What is the chromosome number in a body cell of this animal?

A. 3 **B.** 6

C. 12 **D.** 24

5. Which of the following diagrams correctly shows a cell of the same animal at metaphase of the first meiotic cell division?

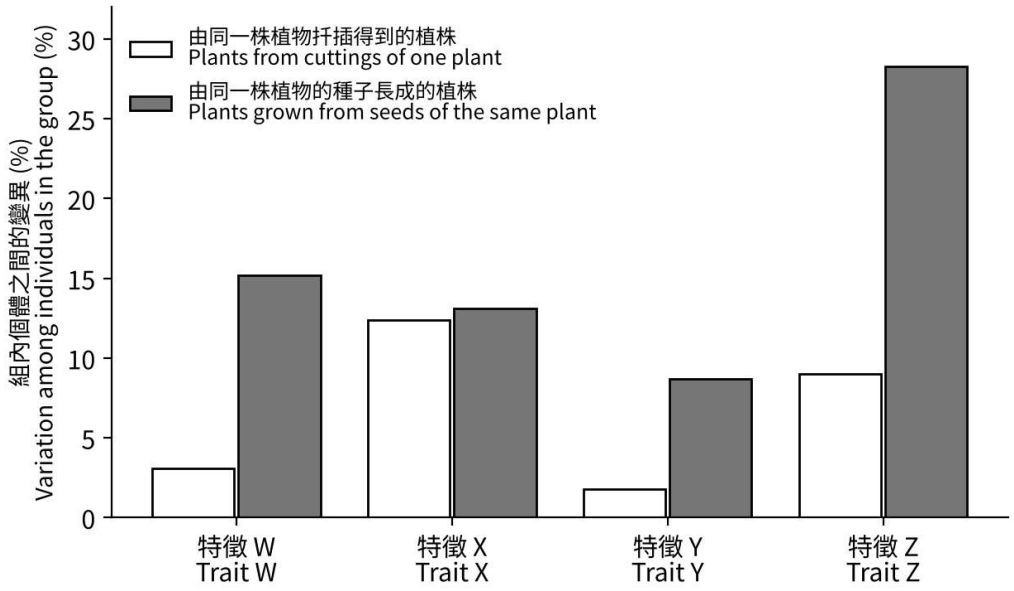


虛線表示細胞的赤道；每條染色體由兩條染色單體組成
Dotted line: equator of the cell. Each chromosome consists of two chromatids.

A. A
B. B
C. C
D. D

6. One group of plants was grown from cuttings of a single plant, and another group was grown from seeds of the same plant. Both groups were grown in the same field. The variation among individuals within each group was measured for four traits W, X, Y and Z. The results are shown below.

Which trait is most affected by environmental factors?



A. W
B. X
C. Y
D. Z

7. The table below shows the number of differences in the amino acid sequence of protein E among four species K, L, M and N.

With reference to the information above, which of the following can be deduced?

蛋白質 E 的氨基酸序列差異數目
Number of amino acid differences in protein E

	K	L	M	N
K	—	3	12	20
L		—	11	19
M			—	18
N				—

A. N is the oldest and most primitive of the four species.

B. M evolved from K.

C. Protein E of L evolved at the fastest rate.

D. Among the four species, K and L share the most recent common ancestor.

8. After lava cooled on a volcanic island, lichens were the first organisms to grow on the rocks. Dead lichen fragments were collected, sterilised and mixed with sterilised rock particles in two portions. Soil bacteria were added to one portion but not the other. Moss spores were added to both. After a few months, mosses grew only in the portion with soil bacteria.

In the early stage of succession, decomposers help the growth of mosses mainly because they

A. break down dead lichens into carbon dioxide for photosynthesis of mosses.

B. convert nitrogen in the atmosphere into nitrates for absorption by mosses.

C. break down dead organic matter to release minerals, forming soil with the rock particles.

D. remove dead lichens to make space for mosses to grow.

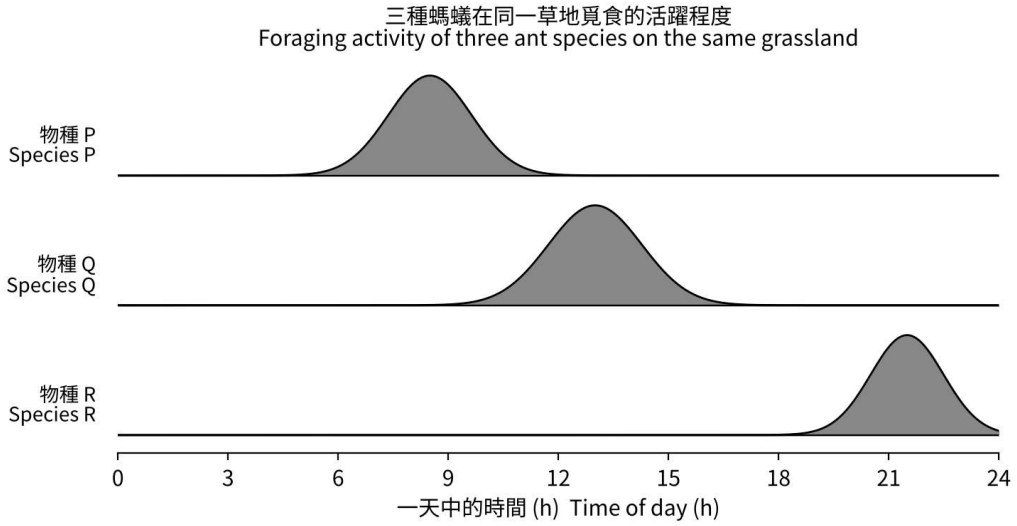
9. The diagram below shows the foraging activity of three ant species P, Q and R on the same grassland. All three species feed on the same kinds of small insects.

Which of the following can be deduced from the above information?

(1) The three species occupy different ecological niches.

(2) Interspecific competition between species Q and R is likely to be weaker than that between P and Q.

(3) The food of these ants is available only during the periods when they are active.



A. (1) only

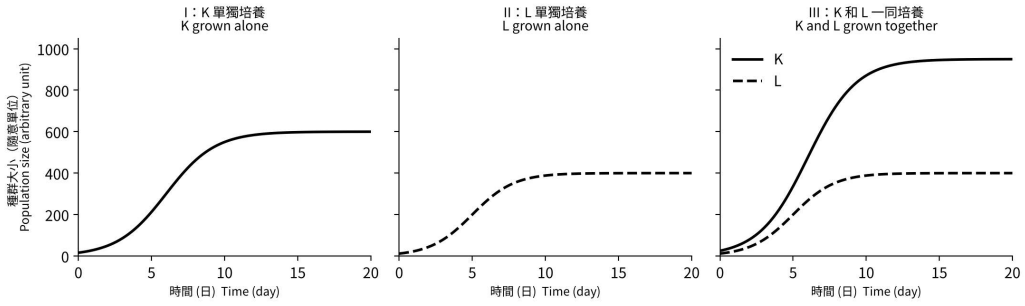
B. (2) and (3) only

C. (1) and (2) only

D. (1), (2) and (3)

10. Fungus K and bacterium L were grown alone and together under the same conditions. The results are shown below.

What is the most likely relationship between K and L?



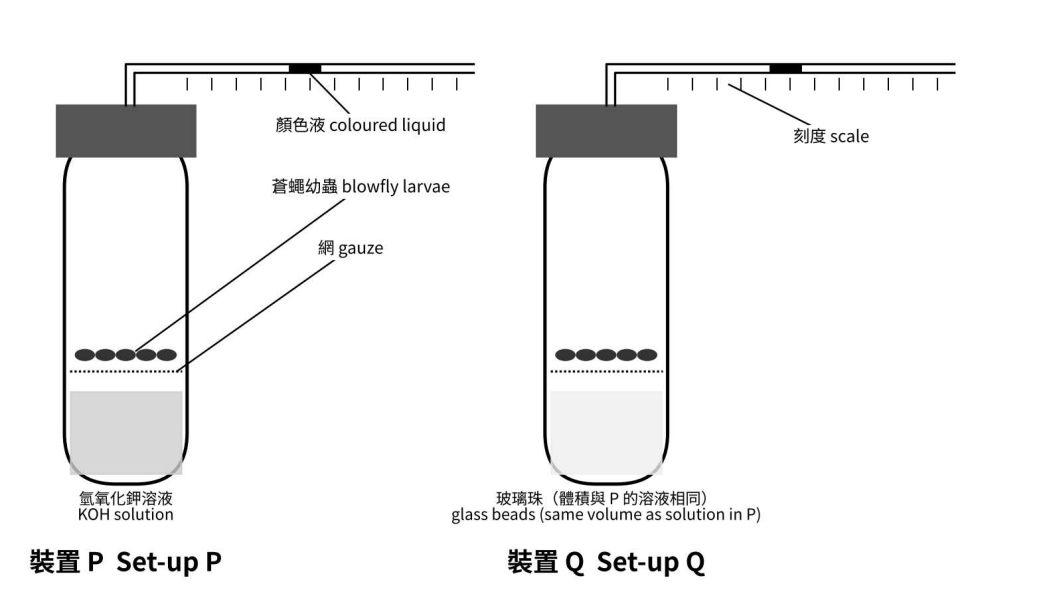
A. competition

B. mutualism

C. parasitism (K is the parasite)

D. commensalism

Directions: Questions 11 and 12 refer to an experiment in which a student used set-ups P and Q below to study respiration of blowfly larvae. Both set-ups were placed in a water bath at 25°C. The glass beads in set-up Q made the volume of air in both set-ups the same.



11. Assume that the blowfly larvae use only glucose as the respiratory substrate and carry out aerobic respiration only. What will happen to the coloured liquid in set-ups P and Q after 10 minutes?

A. P: moves towards the tube; Q: stays still

B. P: moves towards the tube; Q: moves towards the tube

C. P: moves away from the tube; Q: stays still

D. P: stays still; Q: moves away from the tube

12. When the experiment was actually carried out, the student found that the coloured liquid in set-up Q moved slowly towards the tube by 3 mm in 10 minutes. In aerobic respiration, the volume of carbon dioxide released per volume of oxygen used is 1.0 for carbohydrate and 0.7 for fat.

Which of the following best explains this result?

A. The blowfly larvae also carried out anaerobic respiration.

B. The temperature of the water bath rose during the experiment.

C. The glass beads absorbed the carbon dioxide released by the larvae.

D. The blowfly larvae also used fat as a respiratory substrate, taking in more oxygen than the carbon dioxide released.

Directions: Questions 13 and 14 refer to the table below. A student added four strains of yeast separately to equal volumes of the same glucose solution and measured the volume of carbon dioxide released in 30 minutes under anaerobic conditions. The results are shown below.

酵母品系 Yeast strain	30 分鐘內釋出的二氧化碳體積 (cm^3) Volume of CO_2 released in 30 min (cm^3)
K	12.4
L	8.1
M	15.7
N	3.2

13. The student concluded that strain M has the highest rate of anaerobic respiration. Which of the following changes would best ensure that the comparison among the four strains is a fair test?

A. Repeat the experiment three times for each strain and calculate the mean.

B. Use the same number of living yeast cells for each strain.

C. Present the results as a bar chart.

D. Extend the measuring time to 60 minutes.

14. In this investigation, how is ethanol formed in the yeast cells?

A. converted from pyruvate in the mitochondria

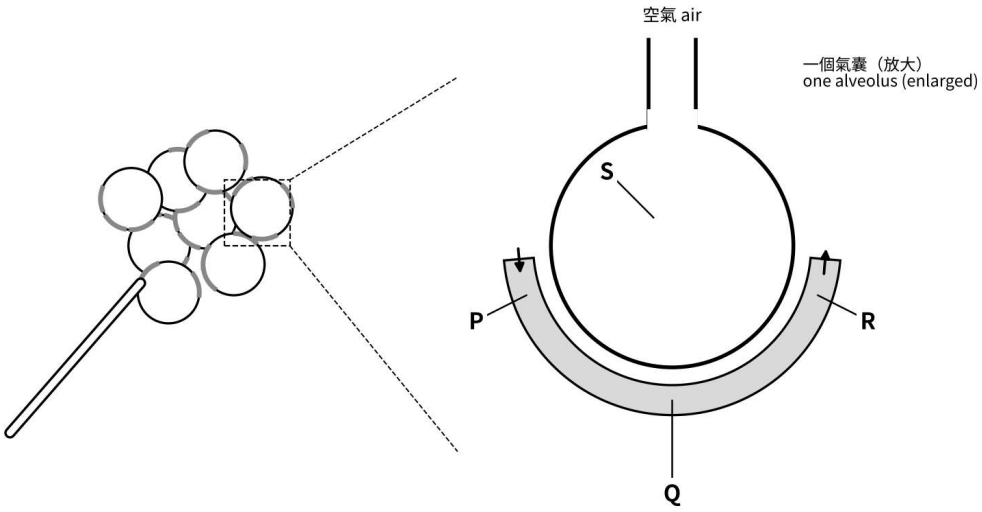
B. formed directly from the breakdown of glucose in the cytoplasm

C. converted from lactic acid in the mitochondria

D. converted from pyruvate in the cytoplasm

15. The diagram below shows a group of alveoli in a human lung, and one enlarged alveolus with the blood capillary around it.

Which labelled part has the highest concentration of carbon dioxide?



灰色帶為微血管；箭咀示血流方向 Grey band: blood capillary; arrows: direction of blood flow

A. P

B. Q

C. R

D. S

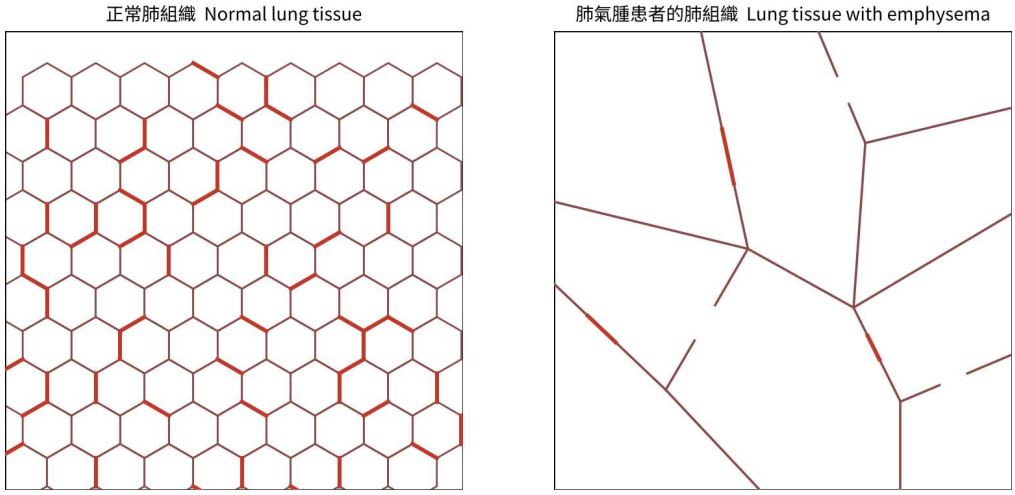
16. In people with emphysema, the walls of the alveoli are damaged and neighbouring alveoli merge. The lung tissue also loses elasticity. The diagram below shows normal lung tissue and lung tissue with emphysema.

Which of the following are likely to occur in a person with emphysema?

(1) The total surface area for gas exchange in the lungs decreases.

(2) The diffusion distance between the air in the alveoli and the blood is shortened.

(3) More air remains in the lungs after each exhalation, so the oxygen concentration gradient between the alveoli and the blood becomes smaller.



兩圖放大倍數相同；細線＝氣囊壁，粗紅線＝微血管，空白處＝氣囊內的空間
Same magnification. Thin lines: alveolar walls; thick red lines: blood capillaries; white: air spaces

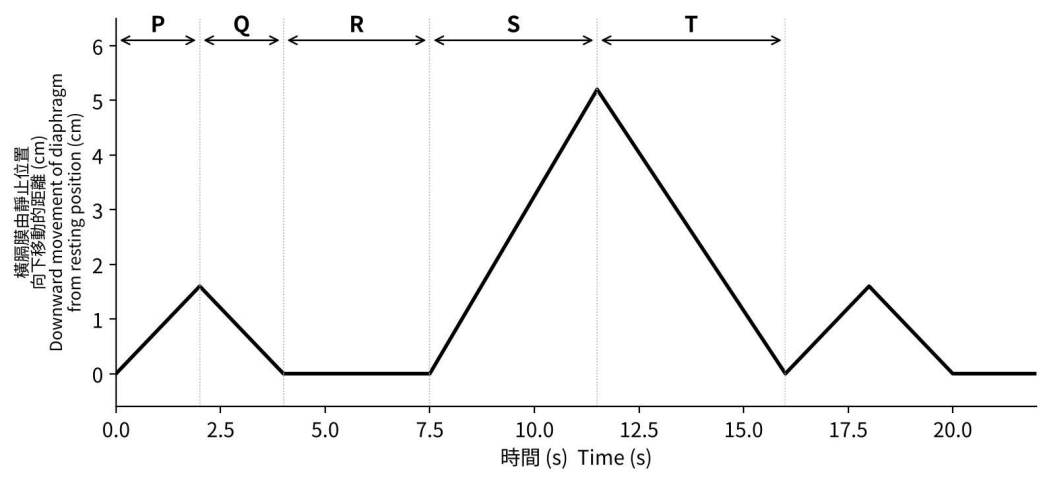
A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

Directions: Questions 17 and 18 refer to the graph below. The position of the diaphragm of a man sitting at rest was measured using ultrasound. The graph shows the downward movement of his diaphragm from its resting position.



17. At which of the following moments was the air pressure inside the lungs of the man equal to the atmospheric pressure?

A. 1 s
B. 3 s
C. 5 s
D. 10 s

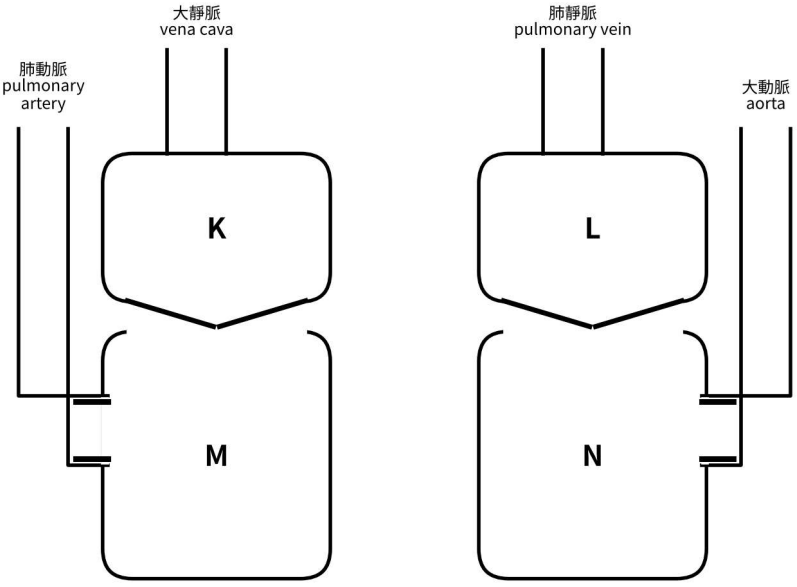
18. Which of the following descriptions about period S are correct?

(1) The muscles of the diaphragm contract.
(2) The air pressure inside the lungs is lower than the atmospheric pressure.
(3) The volume of the thoracic cavity decreases.

A. (1) only
B. (2) and (3) only
C. (1) and (2) only
D. (1), (2) and (3)

19. The diagram below shows the heart at a particular moment. The thick black lines show the state of the valves at that moment.

After this moment, the ventricles begin to relax. Which of the following will happen first?



示意圖（前視，未依比例）；粗黑線＝瓣膜 Schematic diagram, front view (not to scale); thick black lines: valves

A. The semilunar valves close.

B. The atrioventricular valves open.

C. The atria begin to contract.

D. Blood flows from the atria into the ventricles.

20. The bicuspid valve (the atrioventricular valve between the left atrium and the left ventricle) of a patient cannot close completely. Which of the following is most likely to be a consequence of this defect?

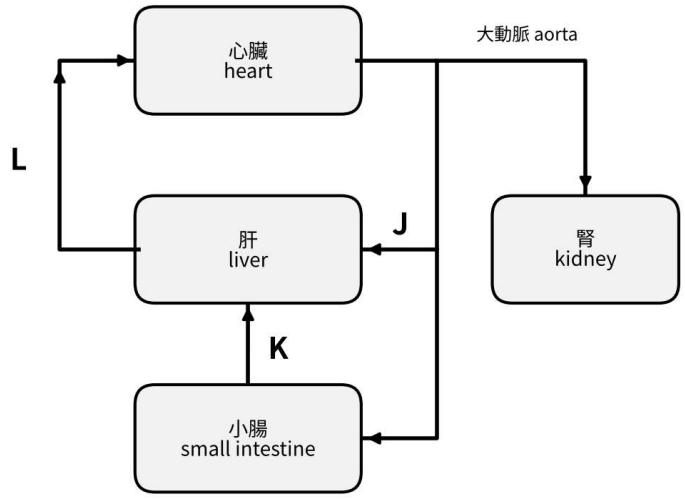
A. When the ventricles relax, blood cannot flow from the left atrium into the left ventricle.

B. When the left ventricle contracts, some blood flows back into the left atrium, so less blood is pumped into the aorta.

C. When the left ventricle contracts, blood flows through the bicuspid valve into the right atrium.

D. When the ventricles relax, blood flows back from the aorta into the left ventricle.

Directions: Questions 21 and 22 refer to the diagram below, which shows part of the human circulatory system. J, K and L represent blood vessels.



箭咀示血流方向（部分血管未有畫出） Arrows: direction of blood flow (some vessels not shown)

21. One hour after a healthy person has taken a protein-rich meal, in which blood vessel is the concentration of amino acids highest, and in which blood vessel is the concentration of urea highest?

A. amino acids: J; urea: K
B. amino acids: K; urea: K
C. amino acids: L; urea: K
D. amino acids: K; urea: L

22. Which of the following descriptions about the urea produced by the liver are correct?

(1) Urea produced by the liver is carried to the kidneys in J.
(2) Urea produced by the liver returns to the right atrium through L and the vena cava.
(3) Urea in L must pass through the pulmonary circulation before it can reach the kidneys.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

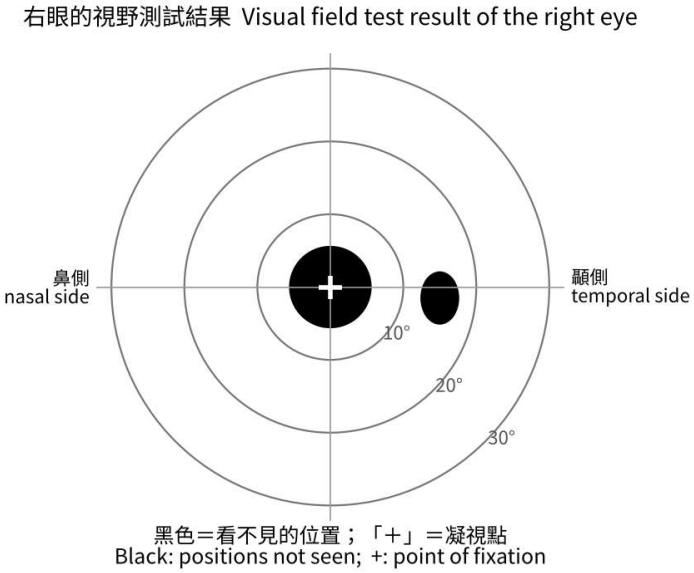
23. Which of the following structures help reduce friction between bones in a movable joint?

(1) cartilage
(2) synovial fluid
(3) ligament

A. (1) only
B. (3) only
C. (2) and (3) only
D. (1) and (2) only

24. A patient looked at the "+" at the centre while small spots of light were shown at different positions to test the visual field of his right eye. The result is shown below.

Which of the following best explains the black area at the centre?



A. The patient's blind spot has enlarged.

B. The patient's lens has become cloudy.

C. The light-sensitive cells at the yellow spot of the patient are damaged.

D. The rod cells at the edge of the retina of the patient are damaged.

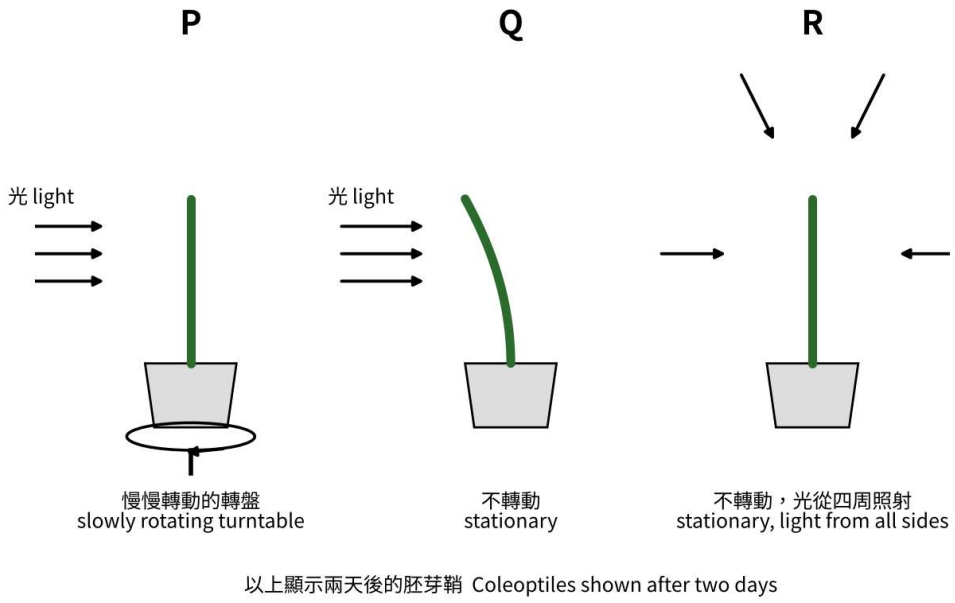
25. An experiment was carried out using three groups of oat coleoptiles. The results after two days are shown below.

Which of the following can be deduced from this experiment?

(1) Coleoptiles bend only when they receive unilateral light.

(2) The tip of the coleoptile is the region that detects light.

(3) In Q, the concentration of auxin is higher on the shaded side of the coleoptile.



以上顯示兩天後的胚芽鞘 Coleoptiles shown after two days

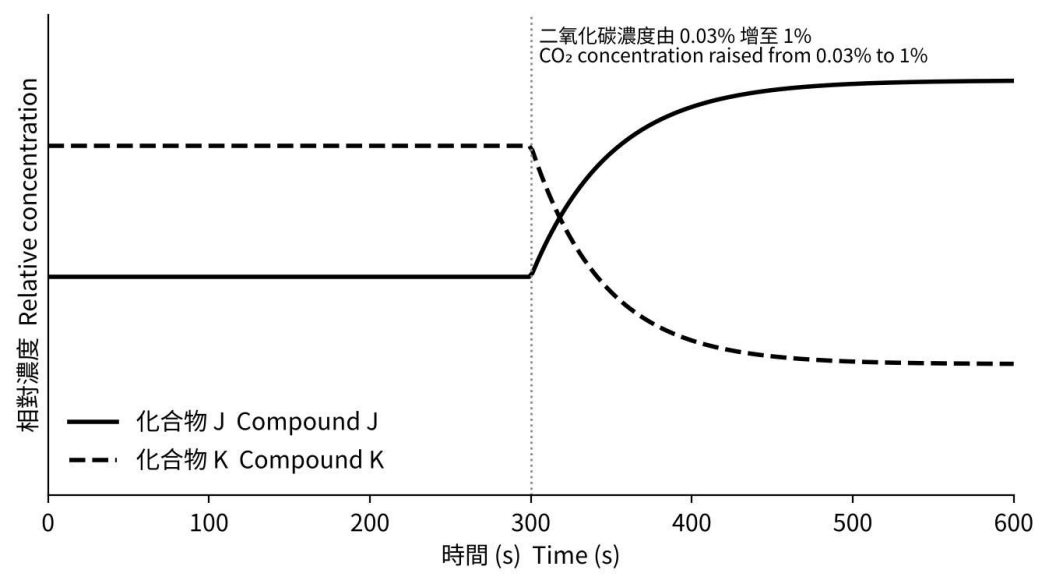
A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

Directions: Questions 26 and 27 refer to an experiment in which spinach leaf discs were placed in a solution under constant light intensity, and the relative concentrations of two compounds J and K in the Calvin cycle were measured. At 300 s, the carbon dioxide concentration of the air bubbled into the solution was raised from 0.03% to 1%. The results are shown below. J and K are the 3-C compound and the 5-C compound, in no particular order.



26. Which of the following combinations correctly shows compound J and the explanation for the change in its concentration?

A. J: 5-C compound; explanation: more 3-C compound is reduced to regenerate 5-C compound

B. J: 3-C compound; explanation: more 5-C compound combines with carbon dioxide to form 3-C compound

C. J: 3-C compound; explanation: more ATP and NADPH are produced in the photochemical reactions

D. J: 5-C compound; explanation: the rate of regeneration of carbon dioxide acceptor increases

27. Which of the following conclusions can be drawn from the results of this experiment?

(1) At 0.03% carbon dioxide, carbon dioxide concentration is a limiting factor of the rate of carbon fixation.

(2) In the first 300 s, the concentrations of J and K remain steady, showing that the rate of formation of each equals its rate of use.

(3) Raising the carbon dioxide concentration increases the rate of the photochemical reactions.

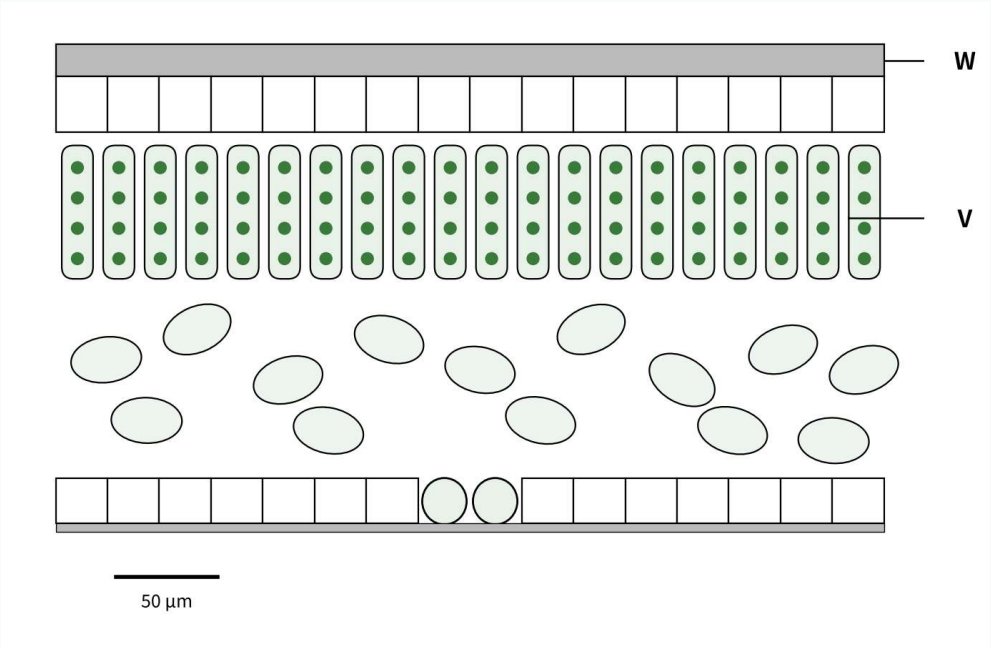
A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

Directions: Questions 28 and 29 refer to the diagram below, which shows a section of a leaf of a plant growing in a dry and sunny area.



28. W is much thicker on the upper surface than on the lower surface of the leaf. Which of the following best explains this feature?

A. There are more stomata on the upper surface, and W prevents water loss through the stomata.

B. The upper surface receives stronger sunlight and is at a higher temperature, and W reduces water loss by evaporation from the upper surface.

C. W consists of dead cells which support the leaf.

D. W stores water for use by the leaf during drought.

29. At a very low light intensity (below the compensation point), what is the direction of the net movement of oxygen and carbon dioxide between cell V and the air spaces in the leaf?

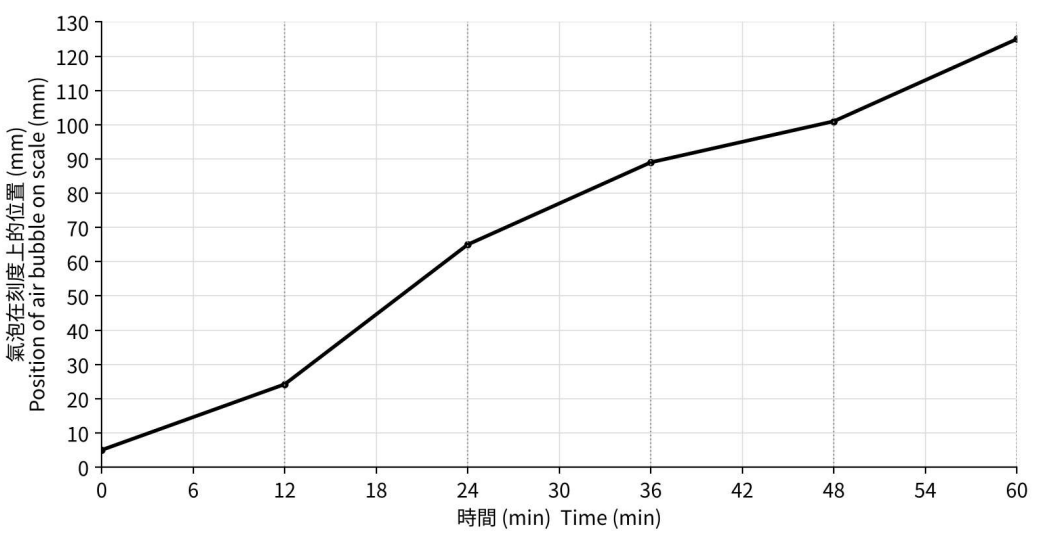
A. oxygen: from V into the air spaces; carbon dioxide: from the air spaces into V

B. oxygen: from V into the air spaces; carbon dioxide: from V into the air spaces

C. oxygen: from the air spaces into V; carbon dioxide: from V into the air spaces

D. oxygen: no net movement; carbon dioxide: no net movement

Directions: Questions 30 and 31 refer to the graph below. A student used a potometer to measure the water uptake of a leafy shoot and recorded the position of the air bubble on the scale. At the start, the temperature was 30°C and the air was still. At 12 min a fan was switched on; at 24 min the temperature was lowered to 20°C (fan still on); at 36 min the fan was switched off; at 48 min the fan was switched on again. The results are shown below. Assume that the rate of water uptake equals the rate of transpiration.



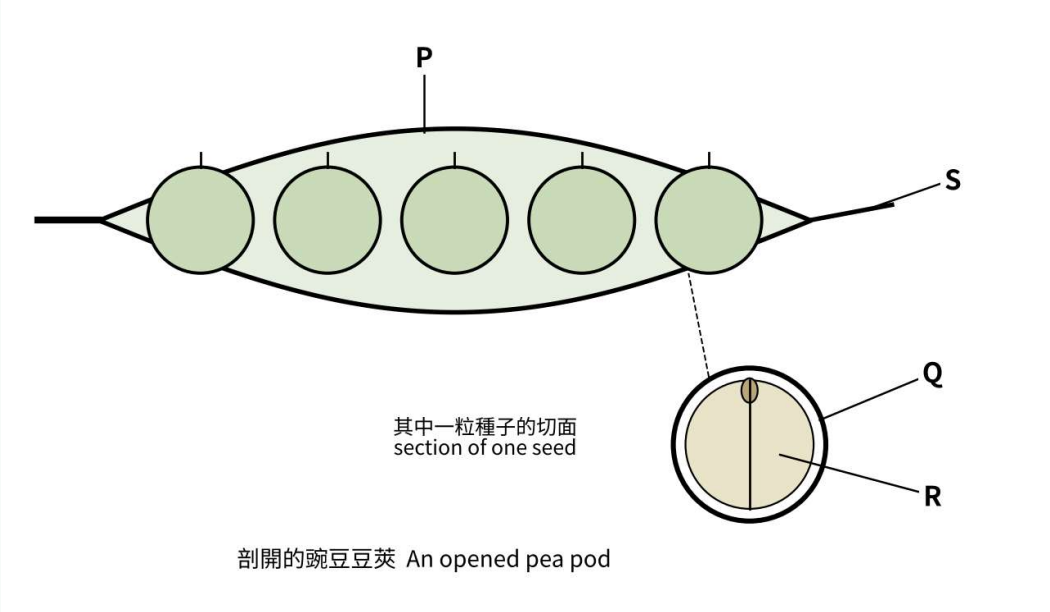
30. During which period is the rate of transpiration of the shoot highest?

A. 0–12 min
B. 12–24 min
C. 24–36 min
D. 48–60 min

31. After which change in conditions does the rate of transpiration fall the most?

A. at 12 min
B. at 24 min
C. at 36 min
D. at 48 min

Directions: Questions 32 to 34 refer to the diagram below, which shows an opened pea pod and a section of one of its seeds. The pea plant bearing this pod has the genotype Aa, and its flower received pollen from a plant of genotype aa.



32. Which of the following structures developed from the zygote?

A. P **B.** Q

C. R **D.** S

33. Which of the following descriptions about the genotypes of structures P, Q and R are correct?

(1) The genotype of P must be Aa.

(2) The genotype of Q may be aa.

(3) The genotype of R may be aa.

A. (1) and (3) only **B.** (1) and (2) only

C. (2) and (3) only **D.** (1), (2) and (3)

34. When the pea matures, P dries, twists and splits open, throwing out the seeds. Compared with seeds dispersed by animals that eat fruits, what is a disadvantage of this method of dispersal?

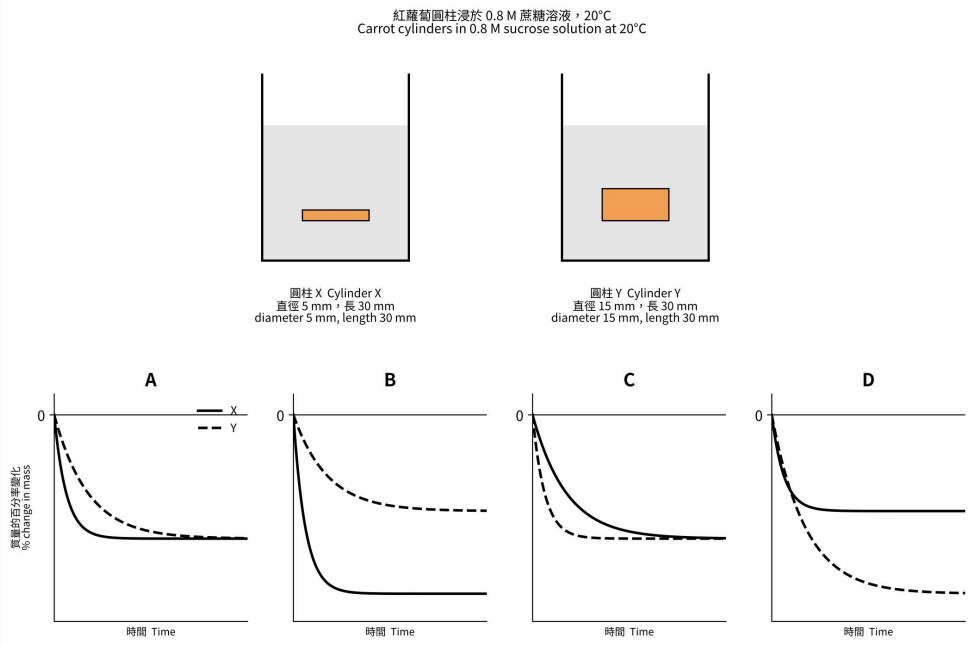
A. The seeds can only be dispersed to places not far from the parent plant.

B. The seeds must pass through the alimentary canal of an animal before they can germinate.

C. The seeds have no testa for protection and easily die from water loss.

D. The seeds lack stored food and grow slowly after germination.

Directions: Questions 35 and 36 refer to the diagram below. A student cut cylinders X and Y of the same length but different diameters from the same carrot. They were placed in 0.8 M sucrose solution at 20°C and their masses were measured at intervals.

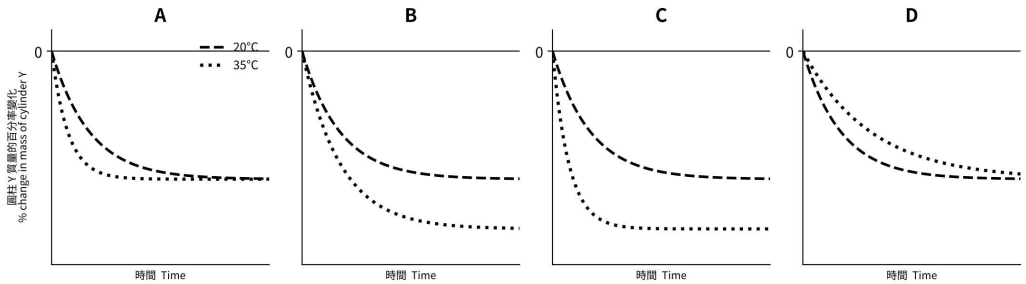


35. Which of the following graphs best shows the percentage change in mass of X and Y?

A. A **B.** B

C. C **D.** D

36. If the experiment is repeated with the same set-up except that the temperature is raised to 35°C, which of the following graphs best shows the percentage change in mass of cylinder Y at 20°C and 35°C?



A. A **B.** B

C. C **D.** D

END OF SECTION A

Section B Structured Questions

1. The table below shows information about three responses in the human body.

反應 Response	刺激後開始出現反應的時間 Time from stimulus to start of response	反應持續的時間 Duration of response
P：氣流吹向眼睛時眨眼 Blinking when a puff of air hits the eye	0.04 s	0.3 s
Q：移居高海拔地區後紅血細胞數目增加 Increase in red blood cell count after moving to high altitude	5 日 days	數月 months
R：飯後血液葡萄糖水平回落 Fall in blood glucose level after a meal	20 min	2 h

(a) Using a table in the order of responses P, Q and R, state whether each response is mainly brought about by nervous coordination or hormonal coordination, and support each answer with one piece of data from the table. (3 marks) (3)

2. Preserved foods such as salted fish contain nitrosamines. Nitrosamines can cause errors in DNA replication, changing some G–C base pairs into A–T base pairs. The rate of nasopharyngeal cancer in Hong Kong used to be high, and this has been linked to frequent consumption of salted fish. The diagram below shows part of the mRNA sequences transcribed from the normal allele and two mutant alleles of a gene. The two mutations were found in the epithelial cells of the nasopharynx of a patient.

mRNA 的部分序列 Part of the mRNA sequence

正常等位基因 Normal allele	AUG	GGC	GAU	…
突變 1 Mutant 1	AUG	AGC	GAU	…
突變 2 Mutant 2	AUG	GGU	GAU	…

密碼子 → 氨基酸
Codon → amino acid

AUG 甲硫氨酸 Met

GGC 甘氨酸 Gly

GGU 甘氨酸 Gly

AGC 絲氨酸 Ser

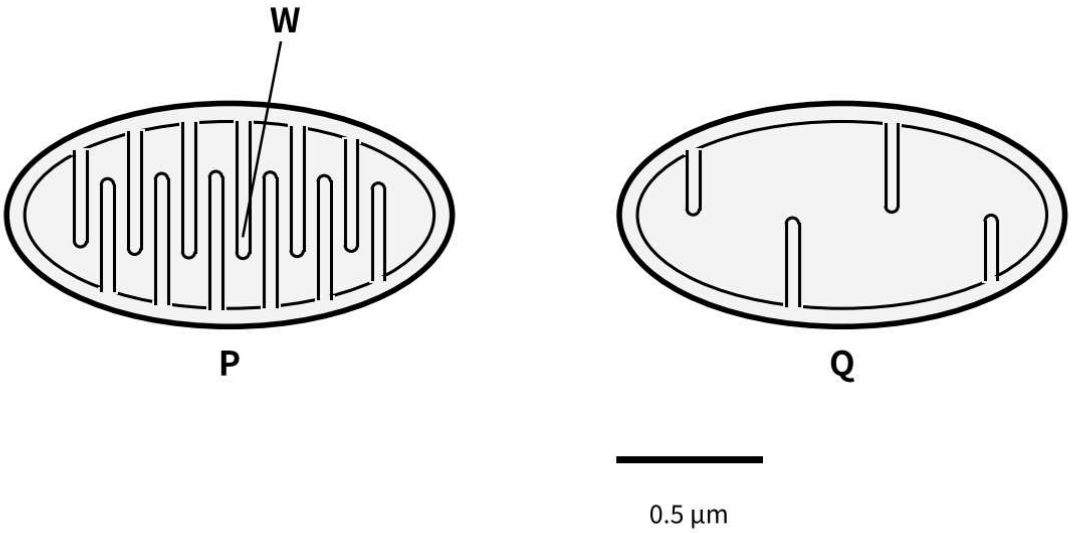
GAU 天冬氨酸 Asp

(a) Based on the change in the base sequence, state the type of gene mutation shown by both mutants. (1 mark) (1)

(b) Explain why the children of this patient will not inherit these two mutations. (1 mark) (1)

(c) The protein product of this gene stops cell division when DNA is damaged. Suggest how mutant 1 may lead to the formation of a tumour. (1 mark) (1)

3. The diagram below is drawn from electron micrographs of two mitochondria. Mitochondrion P was taken from a heart muscle cell and mitochondrion Q from a fat cell.



(a) Name structure W. (1 mark) (1)

(b) With reference to the difference in structure W between P and Q, explain how this difference is related to the functions of the two cells. (2 marks) (2)

(c) The cells of a patient lack the carrier protein that transports pyruvate into the mitochondria.

(i) When glucose is the respiratory substrate, up to which stage of respiration can glucose be broken down in the patient's cells? (1 mark) (1)

(ii) Based on your answer in (c)(i), explain why the concentration of lactic acid in the patient's blood is high. (3 marks) (3)

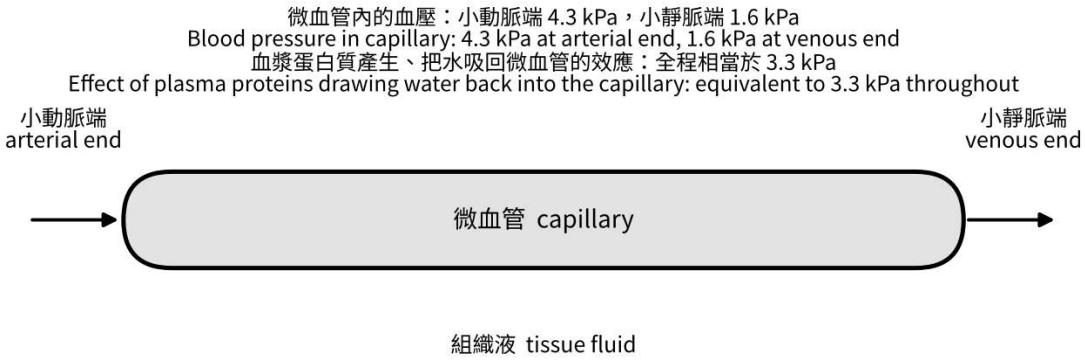
4. The table below shows the levels of some substances in the blood entering and leaving organ Y of a healthy person.

一名健康人士進出器官 Y 的血液樣本（氧和二氧化碳以進入 Y 的血液為 100）
Blood samples entering and leaving organ Y of a healthy person (oxygen and CO_2: blood entering Y = 100)

物質 Substance	進入器官 Y 的血液 Blood entering organ Y	離開器官 Y 的血液 Blood leaving organ Y
氧（相對含量） Oxygen (relative level)	100	28
葡萄糖 (mmol dm^{-3}) Glucose (mmol dm^{-3})	5.0	4.3
乳酸 (mmol dm^{-3}) Lactic acid (mmol dm^{-3})	3.6	7.8
二氧化碳（相對含量） Carbon dioxide (relative level)	100	131

(a) State what organ Y is most likely to be and support your answer with two pieces of evidence from the table. (2 marks) (2)

(b) With reference to the data in the diagram below, explain how tissue fluid is formed at the arterial end of a capillary and how it returns to the capillary at the venous end. (3 marks) (3)



(c) Lymphatic filariasis is caused by parasitic worms blocking the lymph vessels in the legs. Explain why the legs of the patient swell. (3 marks) (3)

5. Protein U is found in the cell membrane of red blood cells. A small number of people are born without protein U, but most of them have no obvious health problems.

Red blood cells from normal people and from people lacking protein U were placed in 2 mol dm^{-3} urea solution, and the percentage of haemolysed red blood cells was measured (Graph I). The two groups of red blood cells were also placed in distilled water, and the percentage of haemolysis was measured over time (Graph II). Blood samples from the same volunteers were used in both experiments.

圖 I Graph I

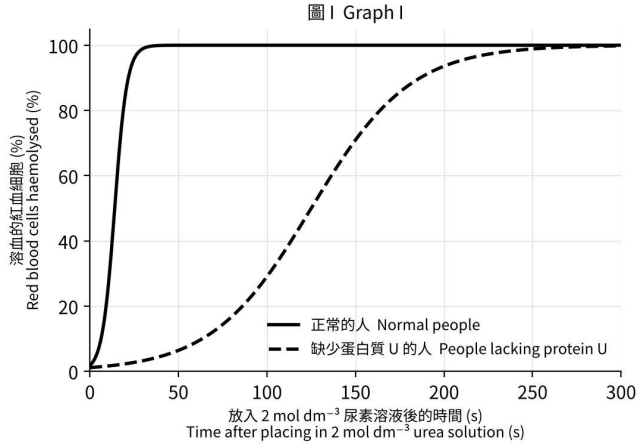
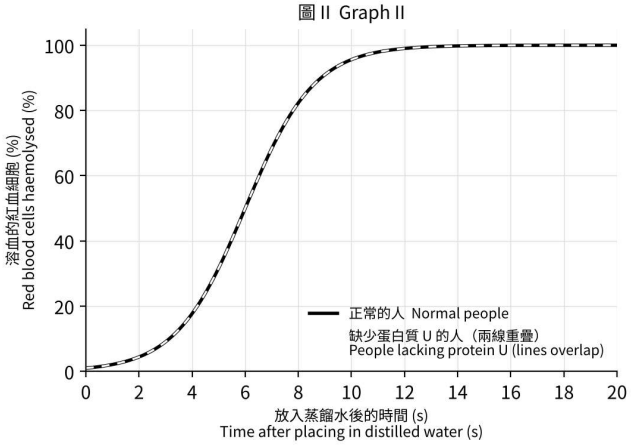


圖 II Graph II



(a) Describe the differences in haemolysis between the two groups of red blood cells in Graph I. (3 marks) (3)

(b)(i) Explain why normal red blood cells are haemolysed in 2 mol dm^{-3} urea solution. (2 marks) (2)

(ii) With reference to Graphs I and II, deduce the function of protein U and support your answer with evidence from both graphs. (3 marks) (3)

(iii) A student said: "These results prove that protein U is a transport protein for urea." State one problem with this statement. (1 mark) (1)

6. Starch-rich foods such as rice are first digested into maltose in the alimentary canal. Maltose is then digested into glucose by maltase. It is claimed that mulberry leaf extract can inhibit maltase.

A student prepared an enzyme extract from the inner lining of a pig small intestine and carried out the following experiment: 2 cm^3 of 2% maltose solution, 1 cm^3 of mulberry leaf extract of different concentrations and 1 cm^3 of enzyme extract were added to each test tube. The tubes were kept at 37°C for 20 minutes and the glucose concentration was then measured. The results are shown in Table 1.

表 1 Table 1

試管 Tube	桑葉提取物濃度 (%) Mulberry leaf extract (%)	酶提取液 Enzyme extract	20 分鐘後的葡萄糖濃度 (mmol dm^{-3}) Glucose after 20 min (mmol dm^{-3})
1	0	有 present	8.4
2	0.25	有 present	6.1
3	0.5	有 present	4.9
4	1.0	有 present	3.3
5	2.0	有 present	3.1
6	0	已煮沸 boiled	0.1

表 2 Table 2：志願者進食白飯後的平均血液葡萄糖水平
Mean blood glucose level of volunteers after a rice meal

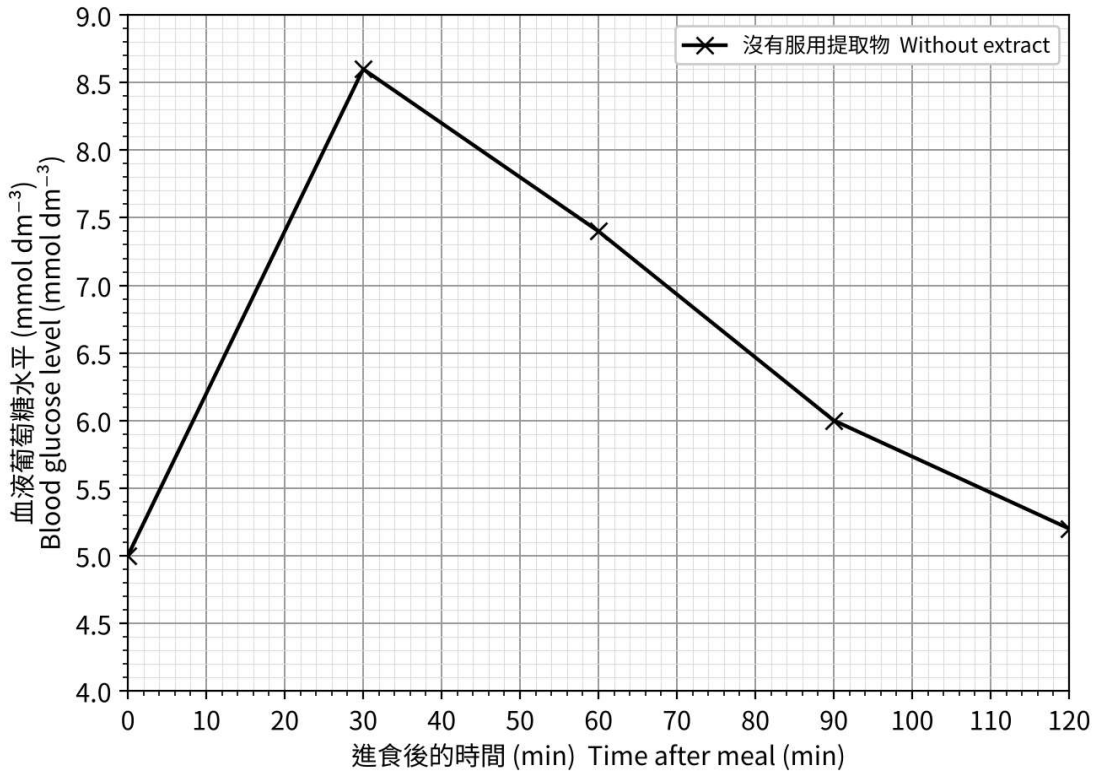
進食後的時間 (min) Time after meal (min)	0	30	60	90	120
沒有服用提取物 (mmol dm^{-3}) Without extract (mmol dm^{-3})	5.0	8.6	7.4	6.0	5.2
服用 1 g 提取物 (mmol dm^{-3}) With 1 g extract (mmol dm^{-3})	5.0	6.9	6.8	6.0	5.3

(a) Explain why inhibiting maltase can reduce the rise in blood glucose level after eating rice. (3)
(3 marks)

(b)(i) Explain why the student used an extract of the inner lining of the small intestine, instead of pancreatic juice, as the source of enzyme. (2 marks) (2)

(ii) The mulberry leaf extract itself may contain a small amount of glucose. Suggest how the experiment could be modified to rule out the effect of this factor on the results. (2 marks) (2)

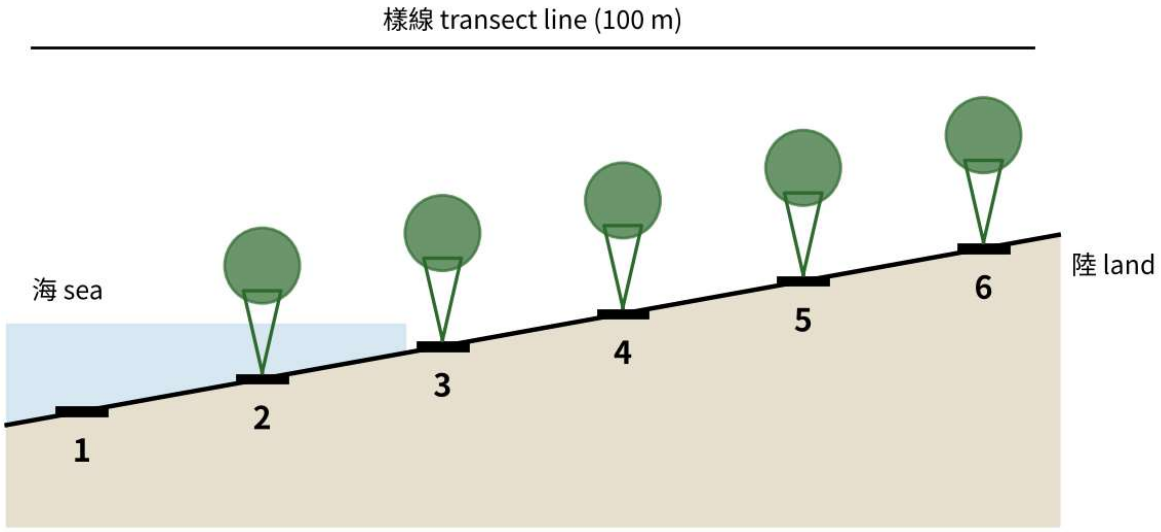
(c) Ten volunteers ate the same amount of rice on two days. On one of the days, they took 1 g of mulberry leaf extract before the meal. Table 2 shows their mean blood glucose levels. On the grid, plot the curve of blood glucose level with the extract. (1 mark) (1)



(d)(i) According to Table 1, 1.0% extract reduced glucose production by about 61%. However, according to Table 2, taking the extract reduced the maximum rise in blood glucose level by only about 47%. Suggest one reason. (1 mark) (1)

(ii) In Table 2, the blood glucose levels of the two groups are almost the same from 90 minutes after the meal onwards. Explain this observation. (2 marks) (2)

7. Students studied the distribution of two snail species S and L in the mangrove at Ting Kok. They laid a 100 m transect line from the seaward side to the landward side and set up six stations along it (diagram below). Graph I shows the density of the two species at each station.



站 1–6：每站放置 5 個 0.5 m × 0.5 m 樣方 Stations 1–6: five 0.5 m × 0.5 m quadrats at each station

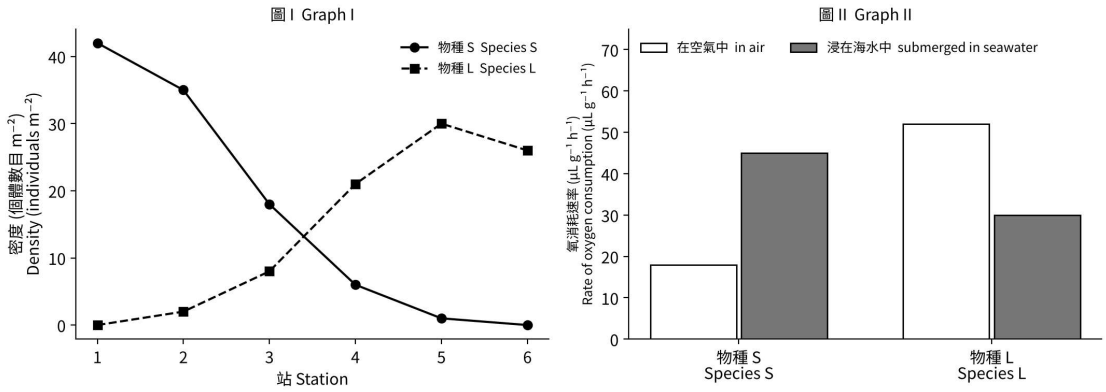
(a)(i) State the name of this sampling method. (1 mark) (1)

(ii) Describe how the density of species L at station 4 can be estimated. (2 marks) (2)

(b) The students hypothesised that species L can still carry out aerobic respiration when exposed to air at low tide. Design an experiment to compare the rate of oxygen consumption of species L in air and when submerged in seawater. (3 marks) (3)

(c) Suggest a result which, if obtained, would support the students' hypothesis. (1 mark) (1)

(d) Graph II shows the results. A student claimed: "Species L is better adapted than species S to being exposed to air, and this explains the distribution of the two species in Graph I." With reference to Graphs I and II and your answer in (c), evaluate this claim. (3 marks) (3)



8. Measles is an infectious disease caused by a virus and transmitted through the air. Children in Hong Kong receive one dose of measles-containing vaccine at the age of 1 and another in Primary 1. In one year, there was a measles outbreak in Hong Kong with 77 cases, 9 of whom had received two doses of vaccine. Graph I shows the age distribution of the cases. Graph II shows the percentage of people with a protective level of antibodies in each age group in a population survey in the same year. The dashed line shows the proportion of protected people needed to stop measles from spreading in the population (95%).

圖 I：一次麻疹爆發的個案
Graph I: Cases in a measles outbreak

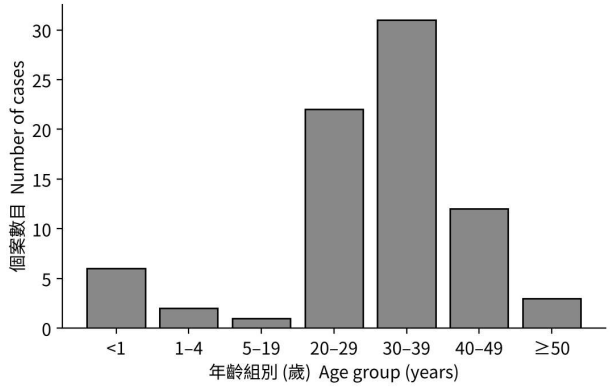
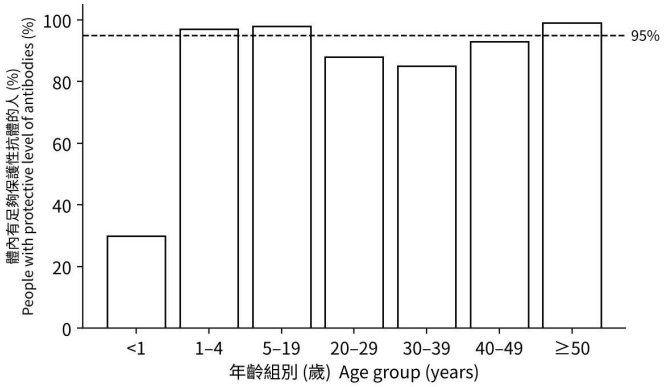


圖 II：同年人口抽樣調查
Graph II: Population survey in the same year



(a) Describe the age distribution of the cases in Graph I. (1 mark) (1)

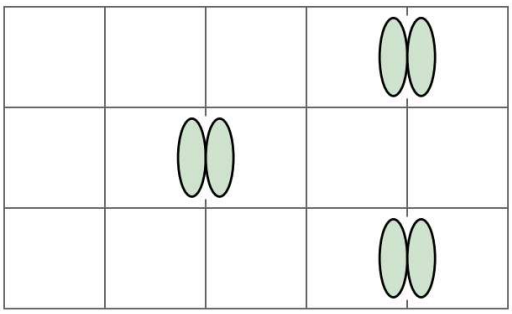
(b) Explain (i) how vaccination gives a person long-term protection, and (ii) why infants who cannot be vaccinated are also protected when a sufficient proportion of the population has protective antibodies. (4 marks) (4)

(c) With reference to Graphs I and II and your answers in (a) and (b), suggest why the outbreak mainly affected adults aged 20–49. (1 mark) (1)

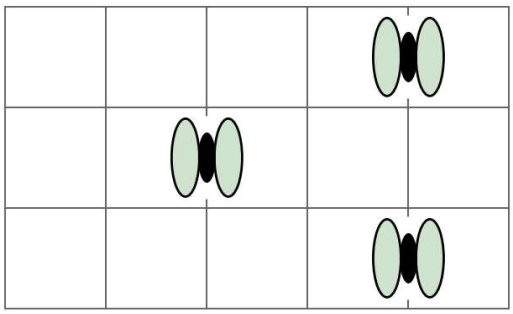
(d) Someone claimed: "Since vaccinated people can still be infected with measles, the measles vaccine is not effective." Based on the information given, evaluate this claim. (2 marks) (2)

9. Fungus F secretes a toxin that keeps the guard cells of leaves turgid. A healthy plant H and a plant I infected by fungus F were both left unwatered. Their lower epidermis was observed and the water content of their leaves was measured. The results are shown below.

停止淋水第 3 天中午的下表皮（示意圖） Lower epidermis at noon on day 3 without watering (diagram)



植物 H（健康） Plant H (healthy)



植物 I（受真菌 F 感染） Plant I (infected by fungus F)

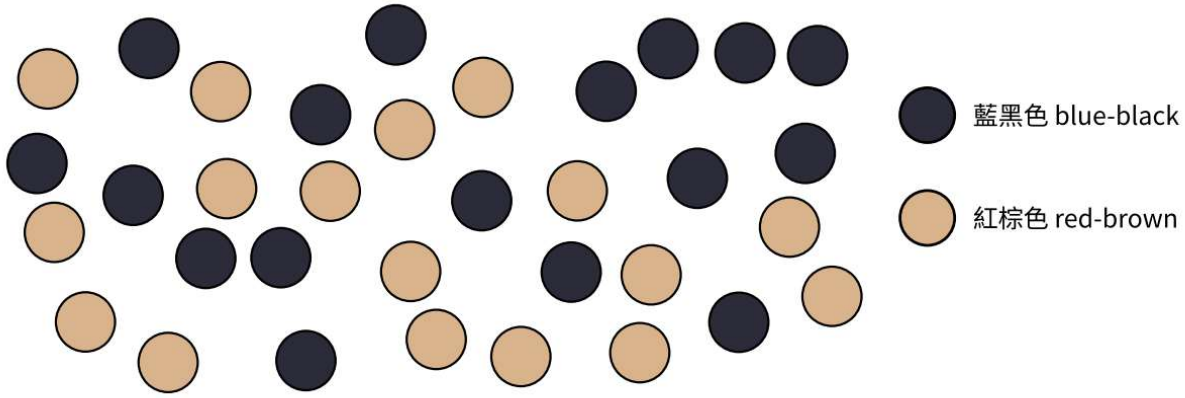
(a) With reference to the lower epidermis of plant I, suggest how keeping the stomata open benefits fungus F. (2 marks) (2)

(b) With reference to the data in the table, explain why plant I wilted a few days after watering stopped. (3 marks) (3)

停止淋水後的日數 Days without watering	0	1	2	3	4
植物 H 葉片含水量 (%) Leaf water content of H (%)	86	85	83	82	80
植物 I 葉片含水量 (%) Leaf water content of I (%)	86	79	70	61	55

10. A gene in rice controls the type of starch. The dominant allele W causes cells to make starch that stains blue-black with iodine solution, while the recessive allele w causes cells to make starch that stains red-brown. Each pollen grain makes starch according to the allele it carries.

Pollen from a rice plant was treated with iodine solution and counted under a microscope: 151 pollen grains were blue-black and 149 were red-brown. The diagram below shows part of the field of view.



以碘液處理後一株稻米植物的花粉（部分視野）
Pollen of one rice plant after adding iodine solution (part of field of view)

(a) Based on the pollen results, use a genetic diagram to deduce the genotype of this plant and show the genotypes and phenotypes of the offspring plants, with their ratios, after self-pollination. (The phenotype of a plant is shown by the colour of its starch with iodine solution.) (4 marks) (4)

(b) The pollen of each blue-black offspring plant was tested with iodine solution. Among the blue-black offspring plants, what proportion produce only blue-black pollen? Explain your answer. (3 marks) (3)

(c) Based on your answer in (a), state the proportion of all the offspring plants that produce blue-black and red-brown pollen in a ratio of about 1:1. (1 mark) (1)

11. You are required to present your answer to the following question in essay form. Criteria for marking will include relevant content, logical presentation and clarity of expression.

In recent years, the numbers of pollinating insects such as bees have declined in many regions, while most of the fruits, vegetables and nuts eaten by humans come from insect-pollinated crops. Discuss the possible effects of the decline in pollinating insects.

(12 marks) (12)

END OF PAPER